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DEPARTMENT OF INFORMATION SCIENCE & ENGINEERING



Network Survey Report on
"NETWORK TOPOLOGY ARCHITECTURE IN BMSIT&M"

Done by,

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ABSTRACT

The data center play an important role in now a day because of the prompt growth of cloud computing. This survey discusses data centers of various organizations and the characteristics and hardware requirements of Data Center Networks (DCN). The service provided, factors affecting the performance and various maintenance issues of DCN's are also discussed. The analysis of our survey can use as an overview of ongoing research in this area. Network security is defined as an activity designed to secure the usability and integrity of the network and information.

Network architecture with its security is a growing concern over the recent years. A campus network faces challenges to address core issues of security which are governed by the network architecture. This report is mainly targeted towards **BMSIT&M campus network** which delivers the required security. This is essential because it prevents the institution from suffering any significant attacks associated with the network.

A university network has a number of uses such as teaching, learning, research, management, e-library, and connections with the external resources. Therefore, network architecture and its security are vital issues for any university.

INTRODUCTION

Internet facility has been crucial in the evolution of our education system in many ways. In order to support its employees and students with best IT infrastructure, BMSIT&M has established one of its the best, robust network infrastructure in 2019-2020 to facilitate reliable and high-speed internet service to both Students, Faculty and Researchers at the campus and BMSIT&M Hostels.

Higher education institutions employ a variety of data centre types to suit their needs, though small data centres, enterprise data centres, and high-performance computing (HPC) are most common. Institutions within the sector are at diverse stages of complete digital transformation, with some already housing sophisticated data centre operations and others running legacy data centre. Implementing cutting-edge technology initiatives can also be a point of competition amongst institutions.

Modern data centres are very different than they were just a short time ago. Infrastructure has shifted from traditional on-premises physical servers to virtual networks that support applications and
workloads across pools of physical infrastructure and into a multicloud environment.

CONSTRAINTS

Air flow

Air flow management addresses the need to improve data center computer cooling efficiency by preventing the recirculation of hot air exhausted from IT equipment and reducing bypass airflow. There are several methods of separating hot and cold airstreams, such as hot/cold aisle containment and in-row cooling units.

Fire protection

Data centers feature fire protection systems, including passive and Active Design elements, as well as implementation of fire prevention programs in

operations. Smoke detectors are usually installed to provide early warning of a fire at its incipient stage.

Although the main room usually does not allow Wet Pipe-based Systems due to the fragile nature of Circuit-boards, there still exist systems that can be used in the rest of the facility or in cold/hot aisle air circulation systems that are closed systems, such as:

- Sprinkler systems
- Misting, using high pressure to create extremely small water droplets, which can be used in sensitive rooms due to the nature of the droplets.

However, there also exist other means to put out fires, especially in Sensitive areas, usually using Gaseous fire suppression, of which Halon gas was the most popular, until the negative effects of producing and using it were discovered.

Security

Physical access is usually restricted. Layered security often starts with fencing, bollards, and mantraps. Video camera surveillance and permanent security guards are almost always present if the data center is large or contains sensitive information. Fingerprint recognition mantraps is starting to be commonplace.

Logging access is required by some data protection regulations; some organizations tightly link this to access control systems. Multiple log entries can occur at the main entrance, entrances to internal rooms, and at equipment cabinets. Access control at cabinets can be integrated with intelligent power distribution units, so that locks are networked through the same appliance.

Energy use

Energy use is a central issue for data centers. Power draw ranges from a few kW for a rack of servers in a closet to several tens of MW for large facilities. Some facilities have power densities more than 100 times that of a typical office building. For higher power density facilities, electricity costs are a dominant operating expense and account for over 10% of the total cost of ownership (TCO) of a data center.

HARDWARE REQUIREMENTS OF DATA CENTERS

- Servers

  Servers are the major component of data center network architecture, which store, analyse and transmit enormous data and directly determine the performance of data center networks



- Switch

  Switches are connecting devices that connects multiple devices together. As far as DCN network is being considered switches are the backbone of the architecture. All switch processing is based on hardware addresses.



- Cable

  Switches, storage devices and servers were connected together with cables, which are the essential components of a data center Network. cables can be coaxial, twisted pairs or fiber optics.

- Storage

  Commonly used storage systems in the data center environment are Network-attached storage (NAS) and Storage area network (SAN). NAS, which is a file-oriented storage network where storage elements are directly connected to Local Area Network. SAN, a high-speed storage network provides superior access to file or block level data for servers.



NETWORK INFRASTRUCTURE

The Internet facility has been crucial to our education system in many ways. In order to support its employees and students with best IT infrastructure, BMSIT&M has established one of its best, robust network infrastructure in 2019 - 2020 to facilitate reliable and high-speed internet service to both Students, Faculty and Researchers at the campus and BMSIT&M Hostels. BMSIT&M has set up its own data centre with all advanced equipment like UTM, Core switch, advanced WIFI Access Points (161nos) optical fibre cabling to interconnect buildings inside campus and Radio Frequency technology to connect Ladies hostel too.

Customer Base and Authentication System:
Total users on board: 3500
Concurrent Users:600-800

Internet user Authentication for Wi-Fi:
Extreme NAC is a centralized authentication system server installed at data centres through which users are created and given to registered users through google form keeping the USN as a unique key for students and employee ID as key for staff.

Bandwidth per User:

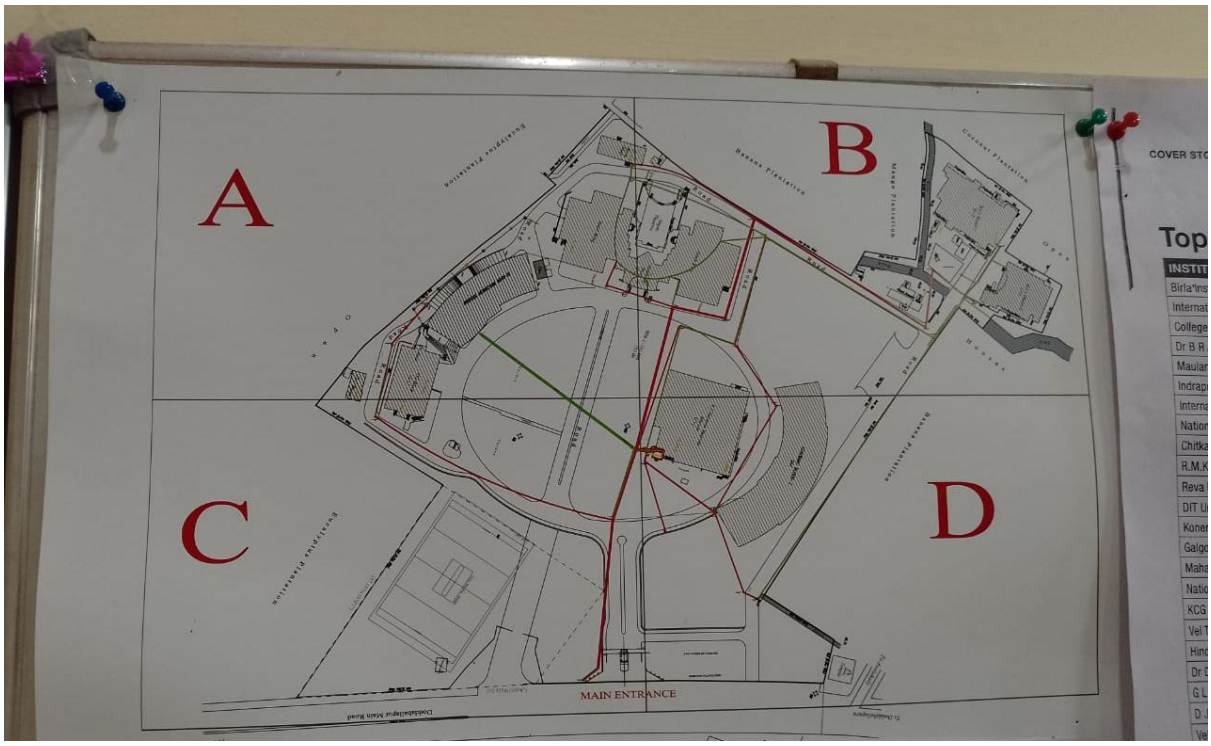
Data per user and bandwidth of 100Mbps upper bandwidth is reserved per user. It is shared as the user increases in the network.

Bandwidth backhaul and distribution:

Band width from two ISPs (350+350=700) Mbps of 1:1 Internet bandwidth ISP links are terminated at data centre and being distributed to the entire campus through optical fibre cable.

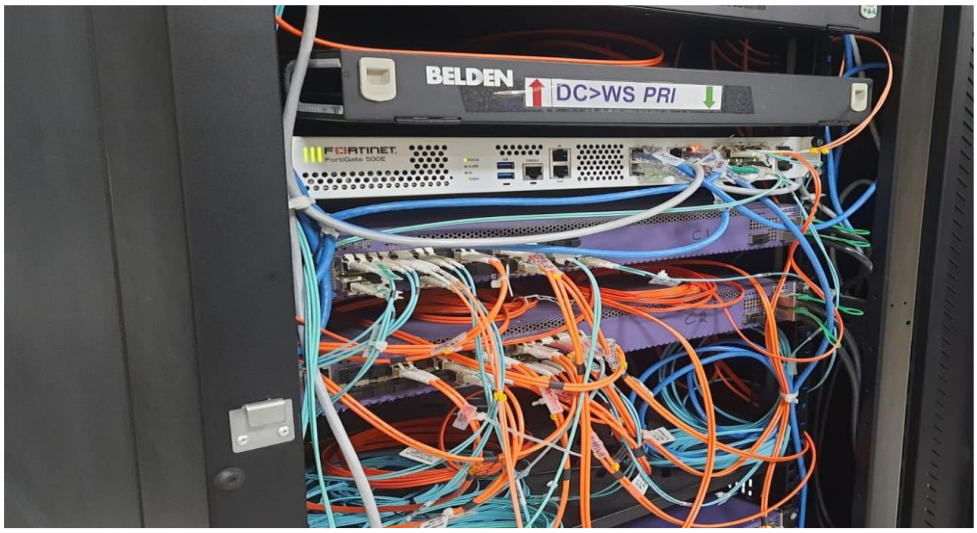
STRUCTURE OF THE NETWORK ARCHITECTURE

The network architecture that has been adopted in BMSIT&M college is a **STAR And Bus based Hybrid Topology**, which has Been widely used in the recent days. In the campus, the Hub originates from the Data Center of the college and is drawn to respective buildings.



Network Distribution:

- Individual requests from Wi-Fi/LAN network connected devices reach the floor switch.
- Multiple floor switches connect at a common node for the block. In fact, BSN block has 2 block nodes (switches) for convenience purposes.
- All of these in turn are connected to their respective stacks as shown below.

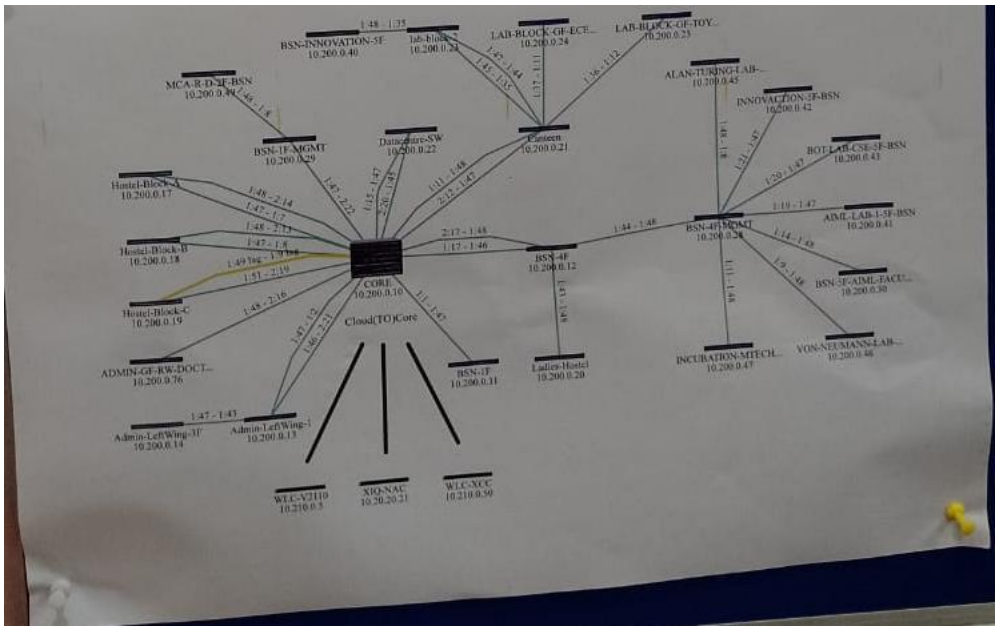


- These stacked nodes are connected to the core switch by using patch cords.
- Usually there is only one core switch required, but in BMSIT we use two core switches (Extreme X590), to accommodate a higher number of incoming connections and higher bandwidths of data exchange, as well as for localizing certain connections to each core switch.
- From the core switch(es), the data enters the UTM (Unified Threat Management) system, where restricted sites are blocked, browsing policies take effect, and cyber-attacks are prevented.
- The next component are the network connectivity modules from **TATA TELE SERVICES** and **SIFY** from where internet access happens, according to the bandwidths.



TOPOLOGY ADOPTED BY COLLEGE

STAR and BUS based Hybrid Topology is implemented in the College.



CONCLUSION

This network architecture survey was primarily aimed at understanding the topology and the types of networking devices that have been used on the campus. We believe that this survey helped us understand and comprehend how important network security is to an organization and we would like to use this knowledge gained through this survey and follow all the norms and protocols to prevent any kind of data breach.